

AI Based Body Measurement App

Final Year Project Report

by

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AI Based Body Measurement App

Executive Summary

*The **AI-Based Body Measurement App** is a mobile application developed to make body measurement more accurate, convenient, and user-friendly by using artificial intelligence. Traditional methods of taking body measurements are often manual, time-consuming, and prone to errors, especially in online shopping, fitness tracking, and tailoring services. This project solves these issues by using **AI-based pose estimation** to automatically calculate body measurements from user-provided images. The system uses **MoveNet and Google ML Kit** for on-device processing, which means all measurements are calculated directly on the user's device without sending images to the cloud. This approach not only improves processing speed but also ensures user privacy and data security. The application is developed using **Flutter** for a smooth and responsive user interface, while **Firebase** is used for authentication and securely storing measurement records.*

*This report brings together the **Project Proposal, Software Requirements Specification (SRS), Software Design Specification (SDS), and the working prototype** of the system. The application allows users to capture or upload images, detect body keypoints using AI, calculate accurate measurements, and store results for future use. The system is designed with a modular and scalable architecture, making it easy to maintain and extend in the future. The functional prototype demonstrates that the system can achieve high measurement accuracy while maintaining full data privacy through local processing. Overall, the project provides a practical and intelligent solution that improves confidence in digital body sizing and has strong potential for future enhancement in areas such as real-time measurement, fitness analysis, and online clothing recommendations.*

FYDP Overview

FYDP Title AI Based Body Measurement App

Sr. No	Roll Numbers	Name	Signatures
1.	BCS22014	Hassan Abdullah	
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Table 1 Project Proposal Summary

FYDP Goals
To achieve highly accurate body measurement extraction using Artificial Intelligence.
To reduce reliance on cloud-based processing and enhance user data privacy through edge computing.
To provide a customized and intelligent online dress design and shopping experience.
FYDP Objectives
Develop an AI-based measurement solution utilizing MoveNet / ML Kit for precise body measurement extraction from static images
Implement model inference and measurement computation locally on the user's device to ensure privacy and real-time processing.
Enable users to browse and select dress designs tailored to their calculated body measurements, ensuring personalized recommendations.
FYDP Success Criteria
Measurement accuracy of almost 90% .
100% data privacy ensured through local (on-device) model execution.
Successful implementation of personalized dress recommendations based on accurate measurements.
Assumptions:
Users possess smartphones equipped with functional cameras and stable internet connections.
Devices support the required OS versions and SDKs for MoveNet / ML Kit pose detection.
Users maintain correct posture and adequate lighting conditions during measurement capture.
Firebase services remain accessible and stable for authentication and backend data management
Risks & Obstacles
Application performance depends on device camera quality and computational capabilities.
Measurement precision may decrease under poor lighting or incorrect posture conditions.
The AI model currently supports single-person body measurement only.
The system does not include payment gateway integration or delivery tracking , which may be considered for future development.
Organization Address: DIT, University of the Punjab, Gujranwala Campus, Pakistan
Target End Users: Online Shoppers, E-commerce Platforms, Tailors, and Fitness Enthusiasts
Suggested Project Supervisor:
Approved By:
Date: November 10, 2025

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Chapter No 1

Project Proposal

1.1.Introduction

This chapter provides an overall introduction to the **AI-Based Body Measurement App** and outlines the motivation behind the project. It explains the background of the problem, the challenges faced by users in traditional body measurement methods, and the need for an automated and intelligent solution. The chapter also defines the problem statement, identifies key stakeholders, and presents the project objectives, scope, assumptions, and potential risks. Together, these sections establish a clear foundation for understanding the purpose and direction of the proposed system.

1.2.Background

Accurate body measurement plays a crucial role in industries such as **online clothing retail, fitness tracking, health monitoring, and tailoring**. Traditionally, body measurements are taken manually using measuring tapes or by visiting physical stores or professionals. These methods are often inconvenient, time-consuming, and prone to human error. With the rapid growth of e-commerce and mobile technology, users increasingly prefer digital solutions that are fast, reliable, and accessible from anywhere.

Recent advancements in **Artificial Intelligence and computer vision** have made it possible to estimate human body measurements using images and pose estimation techniques. AI models such as **MoveNet** and tools like **Google ML Kit** can detect human body keypoints with high accuracy using a smartphone camera. This creates an opportunity to develop a mobile application that can automatically extract body measurements without requiring specialized equipment or professional assistance, while also maintaining user privacy through on-device processing.

1.3.Problem Statement

Despite the growth of online shopping and digital fitness platforms, there is still a lack of reliable and easy-to-use tools for accurate body measurement. Many users face problems such as incorrect clothing sizes, frequent product returns, and inaccurate fitness tracking due to improper measurements. Manual measurement methods depend heavily on user skill and consistency, which often leads to errors.

According to industry reports, **over 30% of online clothing returns occur due to incorrect sizing**, resulting in financial losses for retailers and frustration for customers. Existing digital measurement solutions either require expensive hardware, complex setup procedures, or cloud-based processing that raises privacy concerns. Therefore, there is a strong need for a **low-cost, privacy-preserving, and AI-based mobile solution** that can provide accurate body measurements using a standard smartphone.

1.4.Stakeholders & Interests

The primary stakeholders of this project include:

- **Online Shoppers:** Users who want accurate size recommendations to avoid incorrect purchases. This group represents millions of users globally, with e-commerce fashion users growing rapidly each year.
- **Fitness Enthusiasts:** Individuals tracking body shape and progress over time. The global fitness app market includes hundreds of millions of active users.
- **Tailors and Fashion Designers:** Professionals who require precise measurements for custom clothing without physical presence.
- **E-commerce Businesses:** Companies aiming to reduce return rates and improve customer satisfaction.
- **Developers & Technologists:** IT Professionals Interested in exploring or integrating body measurement features into their own applications, given the app's modular design.
- **End Users (General Public):** Smartphone users interested in quick and convenient body measurement solutions.

Each of these stakeholders benefits from improved accuracy, convenience, and reduced dependency on manual measurement methods.

1.5.Objectives:

The main objectives of the system are:

- To design and develop a **mobile application** for automated body measurement.
- To use **AI-based pose estimation** for extracting body keypoints from images.
- To calculate accurate body measurements using on-device processing.
- To ensure **user privacy** by avoiding cloud-based image processing.
- To provide a simple, user-friendly interface using **Flutter**.
- To store measurement history securely using **Firebase**.

1.6.Scope:

Inclusions:

- Development of a **mobile application**.
- Body measurement extraction from user-captured or uploaded images.
- Use of **MoveNet and ML Kit** for pose detection.
- On-device processing for measurement calculation.
- Secure storage of user data and measurement history.
- Basic size recommendation support.

Limitations:

- The system does not support real-time 3D scanning.
- Accuracy may vary depending on image quality, lighting, and user posture.

- The model is designed for single-person measurement only.
- Does not include payment gateway integration, delivery tracking, or e commerce features.

1.7.Assumptions

The system is designed based on the following assumptions:

- Users will capture images following the provided guidelines.
- The mobile device camera is of sufficient quality.
- The user stands in a proper posture during image capture.
- Internet connectivity is available for authentication and data storage.
- The AI models provide reasonably accurate keypoint detection under normal conditions.

1.8.Risks

The potential risks involved in this project include:

- **Measurement Inaccuracy:** Poor image quality or incorrect posture during body measurement, which may reduce accuracy.
Mitigation: The application Provide on-screen posture guidelines and clear user instructions.
- **Slow performance:** Another risk is slow performance, especially on mid-range or older mobile devices.
Mitigation: This issue is addressed by using optimized and lightweight AI models, such as MoveNet in TensorFlow Lite format, which are designed for real-time execution on mobile hardware.
- **AI Model Limitations:** Pose estimation errors may occur.
Mitigation: Use proven models like MoveNet and conduct extensive testing.
- **User Privacy Concerns:** Handling personal data requires care.
Mitigation: all pose detection and measurement calculations are performed directly on the user's device. No images or video data are uploaded or stored on external servers, ensuring user data remains private and secure. Use on-device processing and secure Firebase authentication.

1.9.Success Criteria

The success of the **AI-Based Body Measurement App** will be evaluated based on the successful implementation and verification of the features defined in the **Software Requirements Specification (SRS)** and **Software Design Specification (SDS)**. The system will be considered acceptable if it fulfills all core functional and non-functional requirements, demonstrates stable performance, ensures user data privacy, and provides a usable and responsive interface.

The following checklist defines the project success criteria.

Table 2 Project Success Criteria

Category	Success Criteria Description	Acceptance Status
User Registration and Authentication	Secure user sign-up, login, and logout using Firebase Authentication	
Core System Functionalities	Image capture or upload, AI-based pose detection using MoveNet and ML Kit	
Body Measurement Calculation	Accurate extraction of body measurements from detected keypoints using on-device processing	
Data Management	Secure storage of user measurement using Firebase	
Reporting and Visualization	Display of measurement results in a clear and understandable format	
Speed and Responsiveness	Efficient processing with minimal latency on supported devices	
Scalability	System architecture supports future feature expansion	
Data Privacy	All image processing performed locally on the device; no cloud image storage	
Access Control	Only authenticated users can access personal measurement data	
User Interface Design	Simple, intuitive, and user-friendly interface developed using Flutter	
Cross-Platform Compatibility	Application runs on supported Android devices	
Error Handling	Proper handling of invalid inputs, failed image capture, and processing errors	
Third-Party Integration	Successful integration of MoveNet, ML Kit, and Firebase services	
Deployment Readiness	System is stable, tested, and ready for academic evaluation and demonstration	

1.10. Tools, Libraries & Technologies

Table 3 Tools Technologies and Libraries

Category	Name	Version	Rationale
Tools	Android Studio	(latest stable version)	Integrated Development Environment (IDE) for building the application, leveraging the Flutter framework.

Tools	Visual Studio Code (VS Code)	(latest stable version)	Lightweight editor for Flutter development, useful for front-end web interface integration.
Tools	Chrome DevTools	(latest stable version)	Essential for debugging and performance testing of the Flutter interface.
Tools	Firebase	(Relevant versions for core services)	Comprehensive backend services for secure user authentication, real-time database, and cloud storage.
Libraries	TensorFlow Lite (TFLite)	^0.10.4 / ^2.0.1	Used for executing the machine learning model locally on the user's device (Edge Computing) to perform real-time body keypoint detection and inference.
Libraries	ML Kit (Google)	(Integrated via SDK)	A preferred model for enhanced on-device pose detection and measurement extraction.
Libraries	Flutter Packages (e.g., camera, image_picker)	(Various versions)	Essential third-party libraries for core functionalities such as real-time camera feed, image handling, and UI enhancements.
Technology	Flutter Framework	v3.x	Cross-platform development

			framework used to build the interactive and responsive app front-end with a unified codebase.
Technology	Edge Computing	N/A	Leverages the user's device's computational capabilities to perform ML inference locally, improving responsiveness, user privacy, and reducing latency.

1.11. Work Division

Table 4 Project Team Members Work Division

Sr. No	Roll Number	Name	Role Assignment & Work Division
1.	BCS22045	M Imran	Project Lead Backend & Database Engineer: Focus on Firebase Integration (Authentication, Real-Time Database, Cloud Storage) and managing data security and logging.
2.	BCS22014	Hassan Abdullah	Frontend & UI/UX Developer: Focus on designing and implementing the User Interface (UI), including image upload, measurement display, and dress visualization.
3.	BCS22044	H M Ertaza	System Developer & AI Model Integrator: Focus on implementing MoveNet based Pose Detection Module (TFLite/ML Kit), algorithms, and ensuring overall system stability.
4.	BCS22046	Dawood Rasheed	QA Specialist & Testing Lead: Focus on Requirements Analysis, designing and executing Test Cases (e.g., Accuracy Validation), and Performance Optimization across devices.

1.12. Conclusion

This chapter presented a comprehensive overview of the **AI-Based Body Measurement App**, establishing the foundation for the proposed system. It introduced the motivation behind the project by highlighting the limitations of traditional body measurement methods and the growing demand for accurate, digital, and privacy-preserving solutions. The background section discussed recent advancements in artificial intelligence and computer vision that enable automated body measurement using mobile devices.

The chapter clearly defined the problem statement, identified key stakeholders, and outlined the objectives, scope, assumptions, risks, and success criteria of the project. By aligning all aspects with the **Software Requirements Specification (SRS)** and **Software Design Specification (SDS)**, this chapter ensures a clear understanding of the system's purpose, boundaries, and expected outcomes. Overall, it provides a structured and well-defined starting point for the detailed analysis, design, and implementation discussed in the subsequent chapters.

Chapter No 2

Literature Review

2.1 Literature Survey

This chapter establishes the foundation for the proposed **AI-Based Body Measurement App** by reviewing and analyzing existing research work and market solutions in the domain of digital body measurement, pose estimation, and clothing size prediction. The literature survey helps in understanding how artificial intelligence, machine learning, and computer vision techniques have been applied to solve body measurement problems in fashion, fitness, and health-related applications.

By critically examining previous studies, this section clearly defines the problem domain addressed by the project. It highlights the limitations of traditional manual measurement methods and identifies the growing need for automated, accurate, and privacy-preserving body measurement solutions that can operate using standard smartphone devices.

2.2 Related Work:

- **Body Shape Classification (2025)**
Supervised ML accurately classifies body shapes and extracts measurements for apparel fitting.
- **ML Clothing Size Prediction (2025)**
Models like SVM and PCA-SVM predict clothing sizes from 3D scans with ~90% accuracy.
- **Fashion Digitization Evaluation (2025)**
3D scanners outperform smartphones for precision tailoring, though phones are improving.
- **3D Body Scanning via Smartphone (2024)**
New 3D avatar reconstruction methods offer accurate body shape and composition data using only a phone camera.
- **Smartphone Body Measurement (2021)**
Smartphones can capture body size measurements with reasonable accuracy, making digital anthropometry affordable.
- **FITME ML Model (2019)**
Machine learning helps estimate body measurements without manual tools.

Market Solutions

Tools like Zozosuit, 3DLook, and Bodygram automate measurements but still struggle with perfect accuracy and e-commerce integration. Their disadvantages are that they rely on manual measurement techniques, which are prone to human error, time consuming, and struggle with seamlessly integrating accurate measurements into personalized clothing recommendations in the e-commerce space.

- **Amazon Body Fit AI** – Size prediction using phone images + purchase data.
- **Walmart Zee kit** – Virtual fitting rooms using ML.
- **Google Body Shape Project** – Phone-based 3D body reconstruction.
- **Sizer Technologies** – Smartphone-based measurement for retail.
- **Bold Metrics** – AI-based clothing size recommendations.

2.3 Gap Analysis:

- **Accuracy and Convenience:** The app aims to provide **highly accurate measurements** (almost 90% accuracy) through automated AI (Move Net/ ML Kit) and a convenient user interface, overcoming the manual techniques that are prone to human error and

inconvenience.

- **Edge Computing/Offline Capability:** The system directly addresses the reliance on cloud processing by implementing **edge computing** for all model execution and measurement calculations locally on the device. This enhances speed, privacy, and allows the core measurement function to operate without continuous internet connectivity.
- **Seamless E-commerce Integration:** The app closes the gap between accurate measurement and informed fashion choice by enabling users to select and visualize dress designs based on their measurements, providing a practical solution to sizing issues and product returns in online shopping.
- **Data Security:** The reliance on local processing and storage ensures the app maintains **100% correctness** in following data privacy policies, a critical gap for sensitive body data.

2.4 Summary

This chapter reviewed existing research, technologies, and commercial solutions related to AI-based body measurement and clothing size prediction. The literature survey highlighted the progress made in machine learning, pose estimation, and smartphone-based measurement systems, while also identifying their limitations in terms of accuracy, privacy, cost, and usability.

The gap analysis demonstrated that there is a strong need for a **low-cost, privacy-preserving, and smartphone-based solution** that performs accurate body measurements using on-device AI processing. These findings directly support the design and development of the proposed **AI-Based Body Measurement App**, which aims to overcome the identified limitations and provide a practical, efficient, and user-friendly solution.

Chapter No 3

Software Requirements

Specification

3.1 Requirements Analysis

This SRS defines the system’s functional and non-functional requirements, identifies key users and use cases, and specifies interface requirements, constraints, system behavior, and requirement traceability.

3.2 User classes and characteristics

Table 5 User classes

User Class	User Characteristics
End User	Uses the mobile application to upload or capture images, view extracted body measurements, receive personalized dress recommendations, and access measurement history. Requires an intuitive and privacy-preserving interface.
System Developer	Responsible for implementing application logic, integrating AI pose estimation models, and ensuring correct interaction between frontend and backend components.
Administrator	Manages the backend system, updates the dress catalog, maintains system configurations, and oversees user data stored in Firebase.
Maintenance Technician	Performs system updates, bug fixes, performance optimization, and ensures continuous system availability.
QA Specialist	Tests the system thoroughly to identify bugs, validate accuracy of measurements, and ensure performance meets requirements.
Security Expert	Evaluates and implements measures to protect sensitive user data such as body measurements and personal images.
Customer Support	Assists end users with troubleshooting, provides help using app features, and gathers feedback for improvements.

3.3 Requirement Identifying Technique

A systematic requirements identification process was adopted to ensure that all system requirements are complete, accurate, and aligned with project objectives. The primary technique used was **Use Case Analysis**, as the system is highly interactive and user-centric. Fully dressed use cases were developed to capture user–system interactions, define system behavior, and derive functional requirements in a structured manner.

In addition, **Storyboarding** was used to visually represent key user workflows such as image capture, measurement display, and dress recommendation. This technique helped validate usability requirements and ensured a logical and user-friendly interaction flow. The combined use

of these techniques enabled clear traceability between user needs, functional requirements, and system behavior.

3.4 Functional Requirements

Use Case Analysis is employed as the primary technique for defining functional requirements due to the highly interactive and user-centric nature of the system.

Functional Requirement-1

Table 6 Function Requirements Specification for User Registration and Login

Identifier	FR-1
Title	User Registration and Login
Requirement	The End User shall be able to register and log in securely using email and password to access personalized features.
Source	Project Proposal / User Needs
Inputs	Email, password
Destination	Mobile application interface
Outputs	Authentication confirmation, user session
Rationale	Required to secure user data and personalize services.
Business Rule	Email must be unique; password length ≥ 8 characters.
Dependencies	None
Priority	High

Functional Requirement-2

Table 7 Function Requirements Specification for Image Capture and Upload

Identifier	FR-2
Title	Image Input Functionality
Requirement	The End User shall be able to capture an image using the device camera or upload an image from the gallery for body measurement processing.

Source	Requirement Analysis
Inputs	Camera image or gallery image
Destination	Temporary system memory
Outputs	Selected image for processing
Rationale	Users must be able to provide up-to-date and accurate images for measurement.
Business Rule	Image must meet minimum resolution and clarity requirements.
Dependencies	FR-1
Priority	High

Functional Requirement-3

Table 8 Function Requirements Specification for Body Keypoint Detection

Identifier	FR-3
Title	AI-Based Pose Detection
Requirement	The system shall detect human body keypoints using MoveNet / ML Kit on-device AI models.
Source	Technical Architecture
Inputs	Image
Destination	Measurement module
Outputs	Set of detected body keypoints
Rationale	Keypoints are required for body measurement calculation.
Business Rule	Images must not be sent to external servers. (Edge computing)
Dependencies	FR-2
Priority	High

Functional Requirement-4

Table 9 Function Requirements Specification for Measurement Calculation

Identifier	FR-4
Title	Measurement Extraction
Requirement	The system shall calculate body measurements such as chest, waist and limb lengths from detected keypoints.
Source	Core System Objective
Inputs	Body keypoints
Destination	Display and storage modules
Outputs	Numerical body measurements
Rationale	Core purpose of the application.
Business Rule	Measurement accuracy must remain within defined tolerance.
Dependencies	FR-3
Priority	High

Functional Requirement-5

Table 10 Function Requirements Specification for Measurement Visualization

Identifier	FR-5
Title	Display Body Measurements
Requirement	The system shall display calculated measurements and detected keypoints visually to the user.
Source	UI Requirements
Inputs	Measurement data
Destination	Mobile screen
Outputs	measurement results

Rationale	Helps users verify correctness of results.
Business Rule	UI elements must not overlap keypoints.
Dependencies	FR-4
Priority	Medium

Functional Requirement-6

Table 11 Function Requirements Specification for Measurement Storage

Identifier	FR-6
Title	Save Measurement History
Requirement	The system shall securely store user body measurements for future access.
Source	User Needs
Inputs	Measurement data
Destination	Firebase database
Outputs	Stored measurement record
Rationale	Enables tracking and reuse of measurements.
Business Rule	Data must be encrypted at rest.
Dependencies	FR-5
Priority	Medium

Functional Requirement-7

Table 12 Function Requirements Specification for Personalized Dress Recommendation

Identifier	FR-7
Title	AI-Based Dress Recommendation

Requirement	The system shall recommend dress designs and sizes based on the user's latest body measurements.
Source	Project Proposal
Inputs	Body measurements
Destination	Recommendation screen
Outputs	Recommended dresses and sizes
Rationale	Required to secure user data and personalize services.
Business Rule	Recommendations follow predefined size-mapping rules.
Dependencies	FR-4
Priority	High

3.5 Non-Functional Requirements

This section specifies nonfunctional requirements other than constraints, supporting requirements recorded in Functional Requirements section, and external interface requirements. These quality requirements should be specific, quantitative, and verifiable.

1. Reliability

The system shall achieve at least **90% measurement accuracy** under standard lighting conditions. The application shall have a **Mean Time Between Failures (MTBF) of at least 30 days** and shall recover from application crashes within **90 seconds** without data loss.

2. Usability

Users shall be able to complete the full body measurement process within **a maximum of five user interactions**. The application interface shall be easy to learn, requiring no more than **10 minutes of initial usage** for a new user to perform core tasks successfully.

3. Performance

The system shall generate body measurement results within **30 seconds** after image submission. User interface response time for navigation actions shall not exceed **2 seconds** under normal operating conditions.

4. Security

All sensitive user data, including images and body measurements, shall be processed on-device or stored in Firebase using **end-to-end encryption**. Unauthorized access attempts shall be logged, and user data shall only be accessible to authenticated users.

3.6 External Interface Requirements

This section provides information to ensure that the system will communicate properly with users and with external hardware or software elements. This section defines how the system interacts with users and external software or hardware components to ensure smooth and reliable communication.

1. User Interfaces Requirements

The application shall follow **Google Material Design guidelines** to ensure consistency and usability. The user interface shall support readable font sizes, appropriate color contrast for accessibility, and intuitive navigation through icons and buttons. Screens include image preview, measurement results, recommendation lists, and history views.

2. Software Interfaces

The system shall integrate with Firebase Authentication for user management, Firestore or Realtime Database for data storage, ML Kit Pose Detection API, and the TensorFlow Lite interpreter for on-device inference.

3. Hardware Interfaces

The system requires a smartphone equipped with a camera, sufficient memory, and processing capability to support on-device AI inference. Optional sensors such as a gyroscope may be used for image orientation handling.

4. Communications Interfaces

An internet connection is required only for Firebase-related operations such as authentication and data synchronization. The body measurement and pose detection modules shall function offline.

3.7 Use case Analysis

1. Use Case #1 (Extract Body Measurements - UC1)

Table 13 Use case Description for the Use case Extract Body Measurements

Field	Description
UC Identifier	UC1
Requirements Traceability	FR-2, FR-3, FR-4, FR-6
Purpose	Extract accurate body measurements from user images using AI pose detection.
Priority	High
Preconditions	User is logged in and camera permission is granted.
Postconditions	Body measurements are calculated, displayed, and stored.
Actors	End User
Extends	None
Main Success Scenario	1. User uploads or captures an image. 2. System detects body keypoints.

	3. Measurements are calculated. 4. Results are displayed to the user.
Alternate Flows	Image quality is insufficient; system prompts user to retry.
Exceptions	Camera failure or model loading error.
Includes	UC2

2. Use Case #2 (Upload Image – UC-2)

Table 14 Use case Description for the Use case Upload Image

Field	Description
UC Identifier	UC2
Requirements Traceability	FR-2
Purpose	Allow user to capture or upload an image.
Priority	High
Preconditions	Camera or gallery access granted.
Postconditions	Image is temporarily stored for processing.
Actors	End User
Extends	None
Main Success Scenario	1. User selects upload option. 2. Camera or gallery opens. 3. User confirms image selection.
Alternate Flows	User cancels the upload.
Exceptions	Permission denied.
Includes	None

3. Use Case #3 (View Dress Recommendations – UC-3)

Table 15 Use case Description for the Use case View Dress Recommendations

Field	Description
UC Identifier	UC3
Requirements Traceability	FR-7
Purpose	Provide personalized dress recommendations and size suggestions based on the user's body measurements.
Priority	Medium
Preconditions	User is logged in and body measurements are available.
Postconditions	Recommended dress list is displayed to the user.
Actors	End User
Extends	None
Main Success Scenario	1. User selects the "View Recommendations" option. 2. System retrieves the latest body measurements. 3. System matches measurements with available dress sizes. 4. Recommended dresses and sizes are displayed.
Alternate Flows	No suitable dress is found; system displays a message indicating no matches.

Exceptions	Database retrieval failure.
Includes	UC1

4. Use Case #4 (Save Measurement History – UC-4)

Table 16 Use case Description for the Use case Save Measurement History

Field	Description
UC Identifier	UC4
Requirements Traceability	FR-6
Purpose	Save extracted body measurements for future reference and reuse.
Priority	Medium
Preconditions	User is logged in and measurements have been generated.
Postconditions	Measurement record is stored securely in the database.
Actors	End User
Extends	None
Main Success Scenario	1. User selects the “Save Measurements” option. 2. System validates measurement data. 3. System stores the data securely.
Alternate Flows	User cancels the save operation.
Exceptions	Database connectivity failure.
Includes	UC1

5. Use Case #5 (View History – UC-5)

Table 17 Use case Description for the Use case View History

Field	Description
UC Identifier	UC5
Requirements Traceability	FR-6
Purpose	Allow the user to view previously saved body measurements.
Priority	Low
Preconditions	User is logged in and measurement history exists.
Postconditions	Measurement history is displayed to the user.
Actors	End User
Extends	None
Main Success Scenario	1. User selects the “View History” option. 2. System retrieves stored measurement records. 3. Measurement history is displayed.
Alternate Flows	No history exists; system displays an informational message.
Exceptions	Network or database access error.
Includes	None

6. Use Case #6 (Admin Updates Dress Catalog – UC-6)

Table 18 Use case Description for the Use case Admin Updates Dress Catalog

Field	Description
UC Identifier	UC6
Requirements Traceability	FR-7
Purpose	Allow the administrator to add, update, or remove dress designs in the catalog.
Priority	Low
Preconditions	Administrator is authenticated.
Postconditions	Dress catalog is updated successfully.
Actors	Administrator
Extends	None
Main Success Scenario	1. Administrator logs into the system. 2. Administrator selects catalog management option. 3. Administrator adds, edits, or deletes a dress entry. 4. System saves the changes.
Alternate Flows	Invalid data entered; system prompts correction.
Exceptions	Permission violation or database failure.
Includes	None

7. Use Case #7 (Notifications and Alerts – UC-7)

Table 19 Use case Description for the Use case Notifications and Alerts

Field	Description
UC Identifier	UC7
Requirements Traceability	FR-1
Purpose	Notify users about important events such as measurement completion or system updates.
Priority	Low
Preconditions	User has enabled notifications.
Postconditions	Notification is delivered to the user.
Actors	End User
Extends	None
Main Success Scenario	1. A system event occurs. 2. System generates a notification. 3. User receives the alert.
Alternate Flows	Notifications are disabled by the user.
Exceptions	Notification delivery failure.
Includes	None

8. Use Case Diagram

The use case diagram illustrates the interactions between external actors and the AI-Based Body Measurement App. Actors include the End User and Administrator, who interact with use cases such as Upload Image, Extract Body Measurements, View Recommendations, Save Measurement History, and Update Dress Catalog. The diagram clearly defines the system boundary and ensures that all user interactions are traceable to functional requirements.

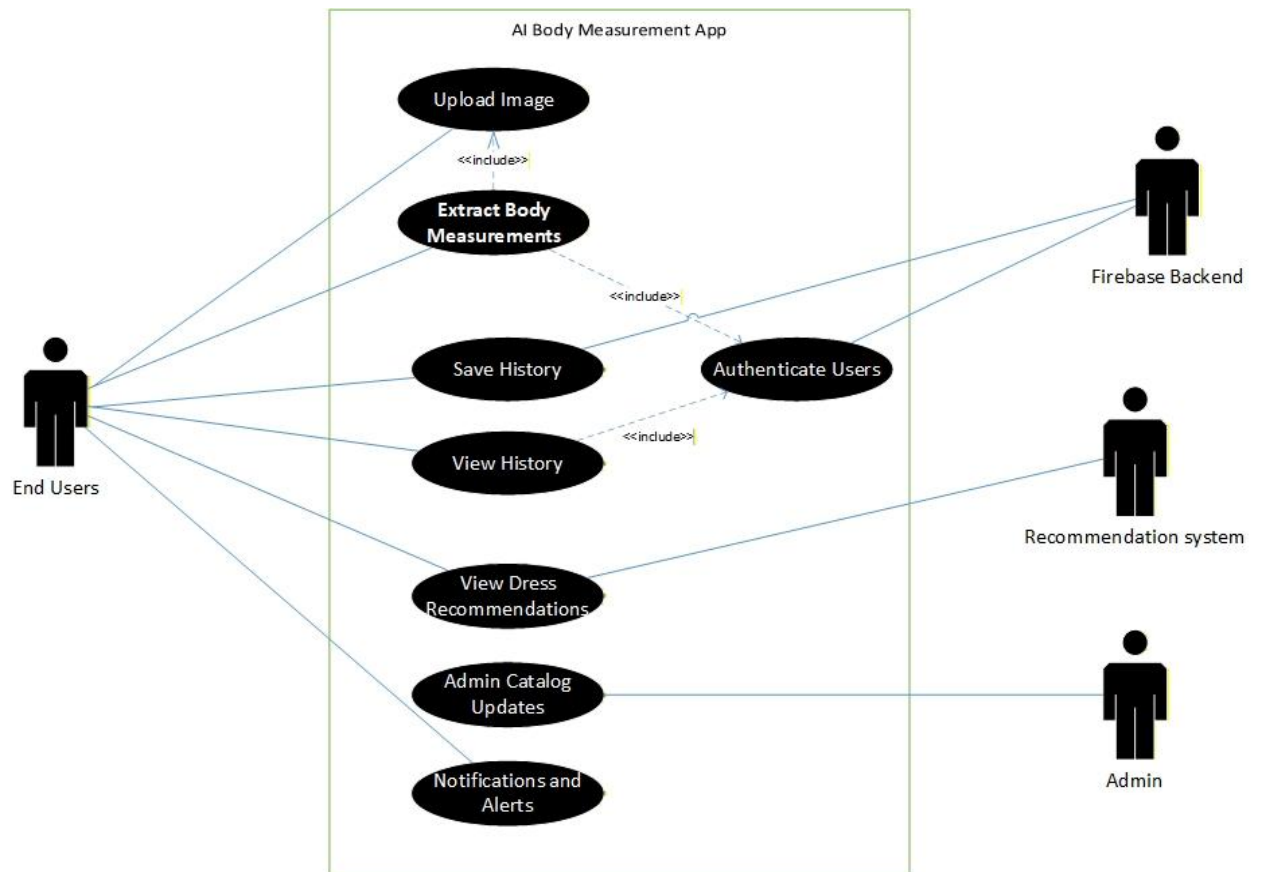
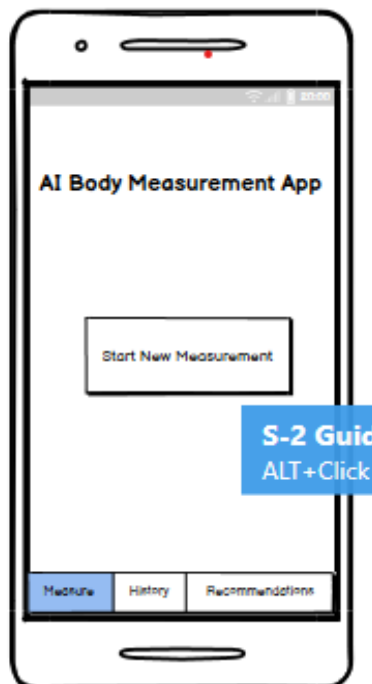


Figure 1 Use Case Diagram

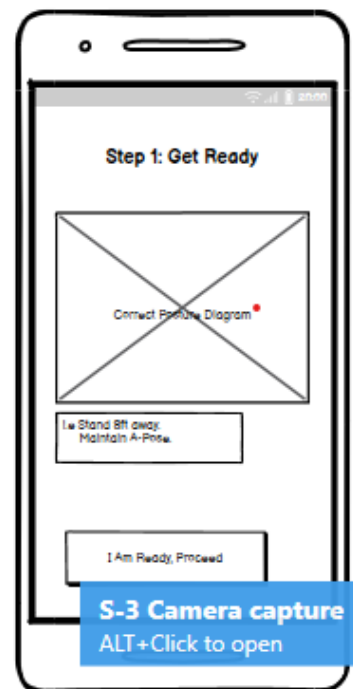
3.8 Storyboards

S-1 Home

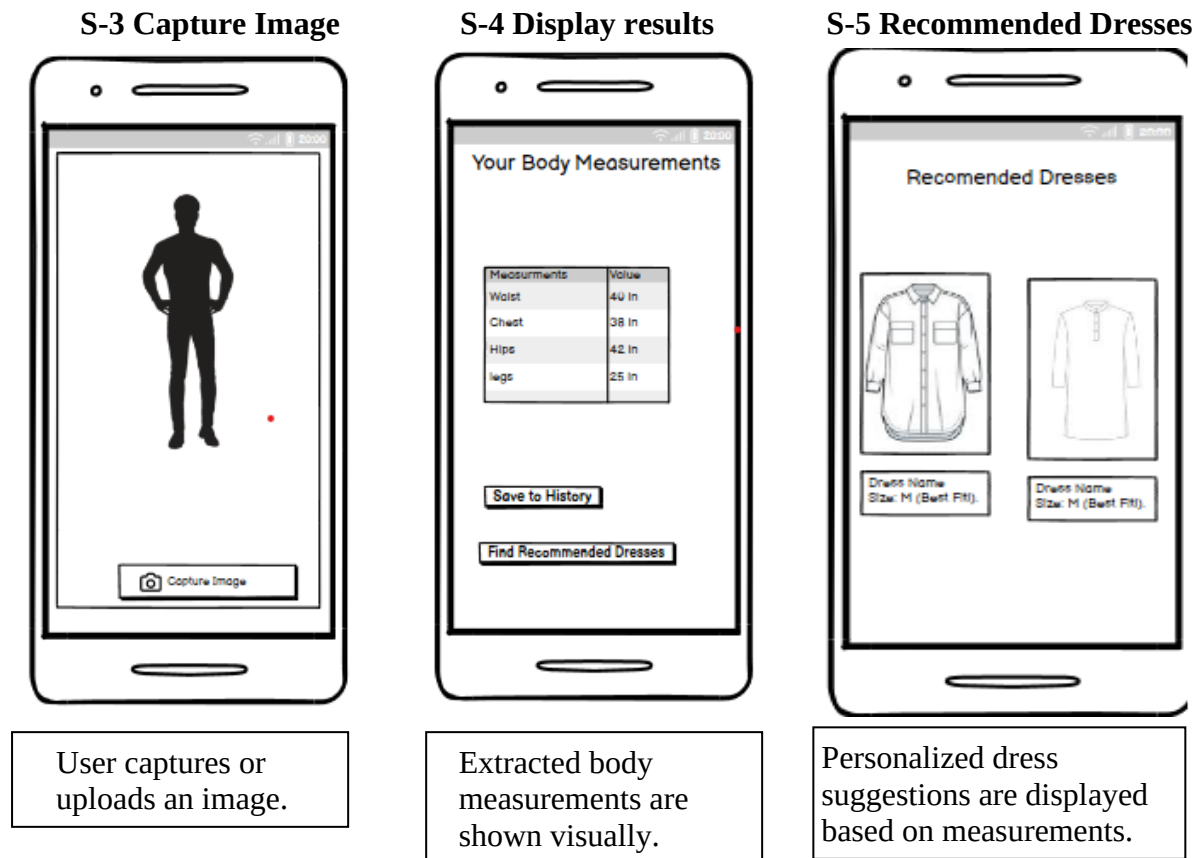


User views main dashboard with options to capture image, view history, or browse recommendations.

S-2 Guidance Screen



Application provides instructions for correct posture and image capture.



3.9 Summary

This System Requirements Specification (SRS) presents a comprehensive and structured definition of the requirements for the AI-Based Body Measurement App. The document establishes a clear understanding of the system’s purpose, scope, functional behavior, non-functional constraints, external interfaces, and user interactions. The primary objective of the system is to accurately extract human body measurements from user images using AI-based pose estimation techniques while ensuring privacy, usability, and reliability.

A systematic requirements engineering process was followed to ensure that the system requirements are complete, consistent, verifiable, and aligned with project goals. The requirements identification phase employed use case analysis and storyboarding techniques to capture user expectations and system behavior effectively. These techniques helped translate high-level project objectives—such as privacy-preserving measurement extraction, personalized dress recommendations, and improved user experience—into well-defined functional and non-functional requirements.

During requirements identification, stakeholders and user classes were analyzed to understand their roles and expectations. End users, administrators, developers, and maintenance personnel were considered to ensure that both operational and technical needs were addressed. Functional requirements were derived by identifying core system operations, including user authentication, image input, body keypoint detection, measurement calculation, storage, and personalized recommendations. Non-functional requirements were defined to enforce system quality attributes such as performance, usability, reliability, and security, which are critical for user trust and system adoption.

Use case analysis played a central role in refining system functionality. Fully dressed use cases were developed to describe user–system interactions in a structured manner, specifying actors, preconditions, postconditions, main success scenarios, alternate flows, and exceptions. This approach ensured that all system behaviors were traceable to functional requirements and supported realistic usage scenarios. Use cases such as “Extract Body Measurements,” “Upload Image,” “View Dress Recommendations,” and “Save Measurement History” provided a clear operational view of the system and guided interface design and implementation planning. The integration of requirements engineering practices ensured that the system design remains aligned with the project’s objectives. By clearly defining requirements early in the development lifecycle, the SRS minimizes ambiguity, reduces development risks, and provides a reliable reference for implementation, testing, and validation. Furthermore, traceability between objectives, requirements, and use cases ensures that every system feature delivers measurable value to users.

In conclusion, this SRS serves as a foundational document that bridges stakeholder expectations and technical implementation. The structured requirements identification process and thorough use case analysis ensure that the AI-Based Body Measurement App is scalable, secure, user-centric, and capable of achieving its intended goals of accurate body measurement extraction, privacy preservation, and enhanced decision-making in fashion selection.

Chapter No 4

Software Design Specification

4.1 System Design

The AI-Based Body Measurement App is a standalone mobile application. Users interact with the system through their smartphone camera to capture body posture. The app processes this input using AI models and displays accurate body measurements in real time.

The system interacts with:

- Mobile device camera
- On-device AI models (MoveNet, ML Kit)
- Firebase Authentication
- Firebase Firestore database

Design Constraints

- The system must run efficiently on mid-range smartphones
- AI processing must be fast and real-time
- Images are not stored or sent to servers for privacy reasons
- Limited mobile device resources (CPU, memory)

4.2 Design Considerations

1. Assumptions and Dependencies:

- Users have a smartphone with a working camera
- Internet is required only for login and saving data
- TensorFlow Lite models are supported on the device

2. Limitations:

- Measurement accuracy depends on lighting conditions
- Loose clothing may affect results
- Only one person can be measured at a time

4.3 Requirements Traceability Matrix

Table 20 Requirements Traceability Matrix

Requirement ID	Scope	Requirement Description	Design Specification
FR-1	User Privacy	The system must allow user registration and secure authentication	FirebaseAuthService, User, AuthUI
FR-2	User Management	The system must capture or upload body images	CameraHandler, ImagePreprocessor
FR-3	Pose Detection	The system must detect body keypoints	PoseDetectionService
FR-4	Measurement Processing	The system must calculate body measurements	MeasurementEngine
FR-5	Results Visualization	The system must display measurement results	MeasurementResultUI

FR-6	Data Persistence	The system must store measurement history	Firestore, MeasurementRepository
FR-7	Recommendation System	The system must provide clothing recommendations	RecommendationEngine, DressCatalogService

4.4 Design Models

The applicable models for the project using object-oriented development approach may include:

1. Design Class Diagram (DCD)

The UML Class Diagram represents the static structure of the AI-Based Body Measurement App. It shows the main classes involved in the system, their attributes, methods, and relationships with each other. This diagram helps understand how system entities are organized and how responsibilities are distributed among different components.

Key classes include User, CameraHandler, PoseEstimator, MeasurementEngine, MeasurementRecord, PoseLandmark, FirebaseService, and RecommendationEngine. The RecommendationEngine is responsible for generating personalized clothing size and design suggestions based on user measurements.

The User class manages user profile and authentication information. The CameraHandler captures images from the mobile device camera. The PoseDetectionService uses MoveNet / ML Kit to detect body keypoints. The MeasurementEngine calculates body measurements such as height, chest, waist, and hip based on detected landmarks. The MeasurementRecord stores calculated values along with timestamps, while FirebaseService handles secure storage and retrieval of user data.

The class diagram provides a clear blueprint of system components and supports an object-oriented design approach, improving modularity, maintainability, and scalability.

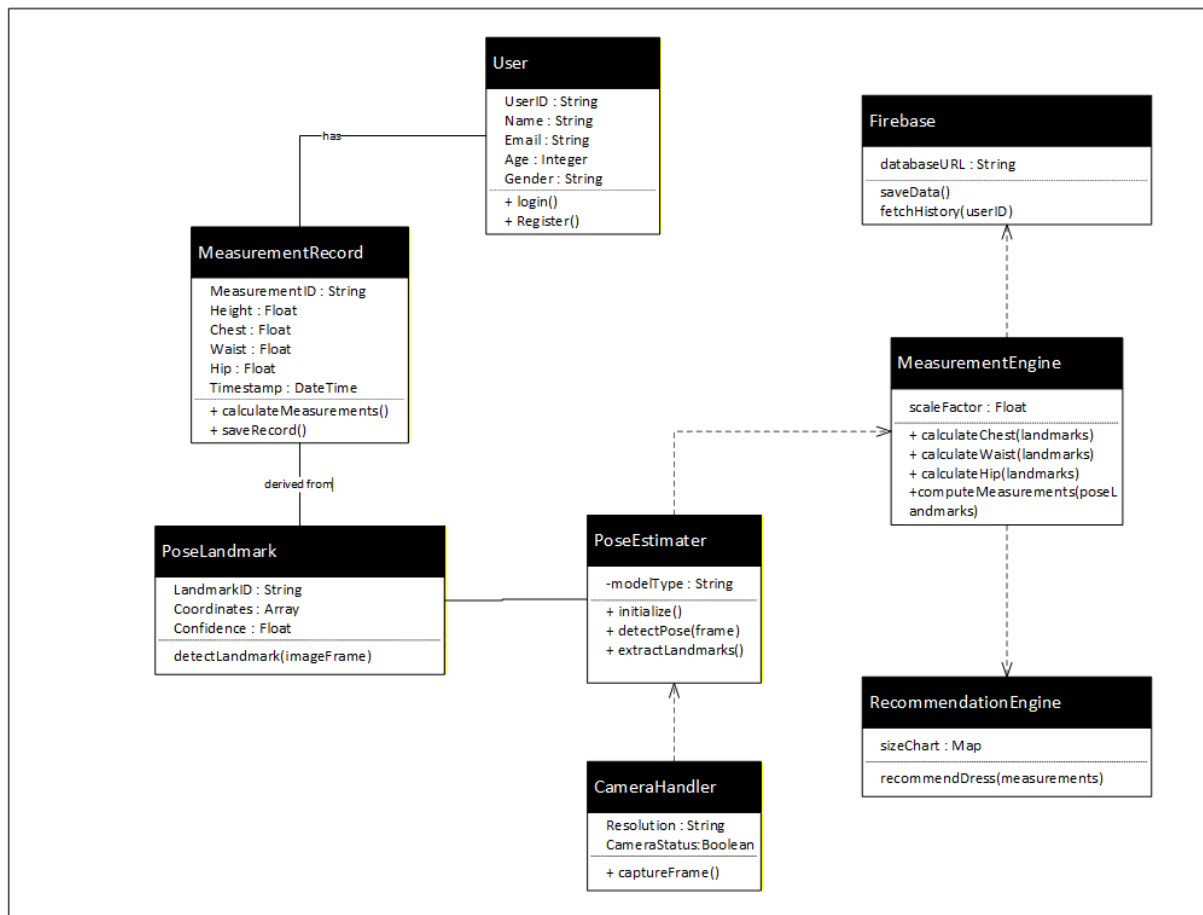


Figure 2 Class Diagram

2. Interaction Diagram (sequence)

The Interaction (Sequence) Diagram illustrates the time-ordered communication between system components during the body measurement process. It focuses on how user actions trigger system responses and how data flows between modules.

This sequence diagram illustrates the interaction between system components during the body measurement extraction process. The user captures or uploads an image through the CameraHandler. The image is processed by the PoseDetector module (MoveNet), which detects body keypoints. These keypoints are then sent to the MeasurementCalculator to compute body dimensions. Finally, the FirebaseService securely stores the calculated measurements in the database.

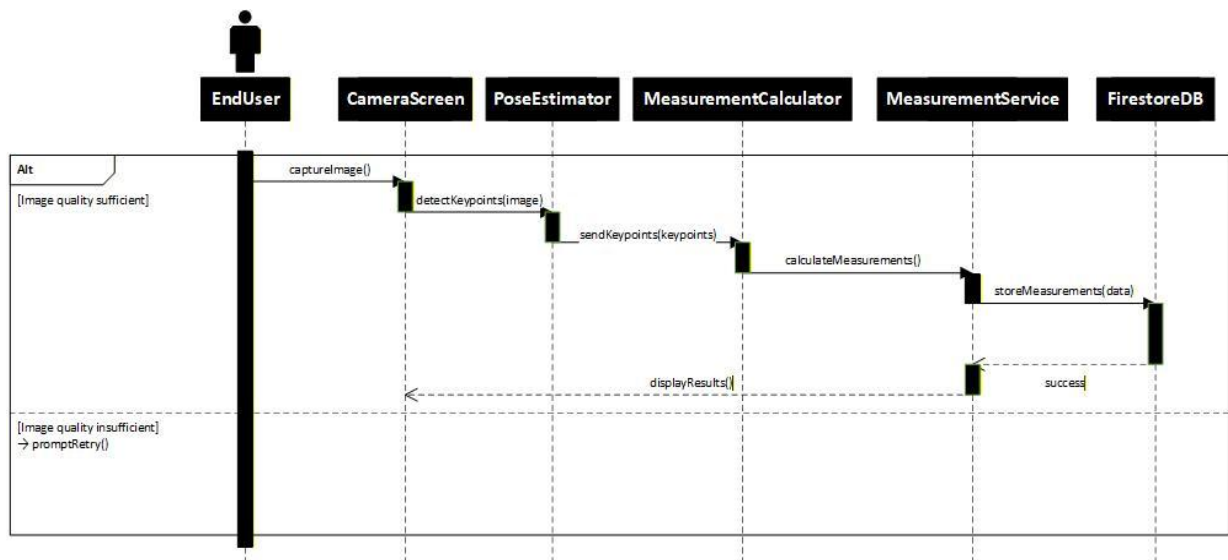


Figure 3 Sequence Diagram for Extract Body Measurements (UC1)

This sequence diagram illustrates the process of uploading or capturing an image. The user selects the upload option, after which the system either opens the camera or gallery based on user selection. Once the image is confirmed, it is temporarily stored for further processing. The diagram also represents alternative flows such as permission denial or user cancellation.

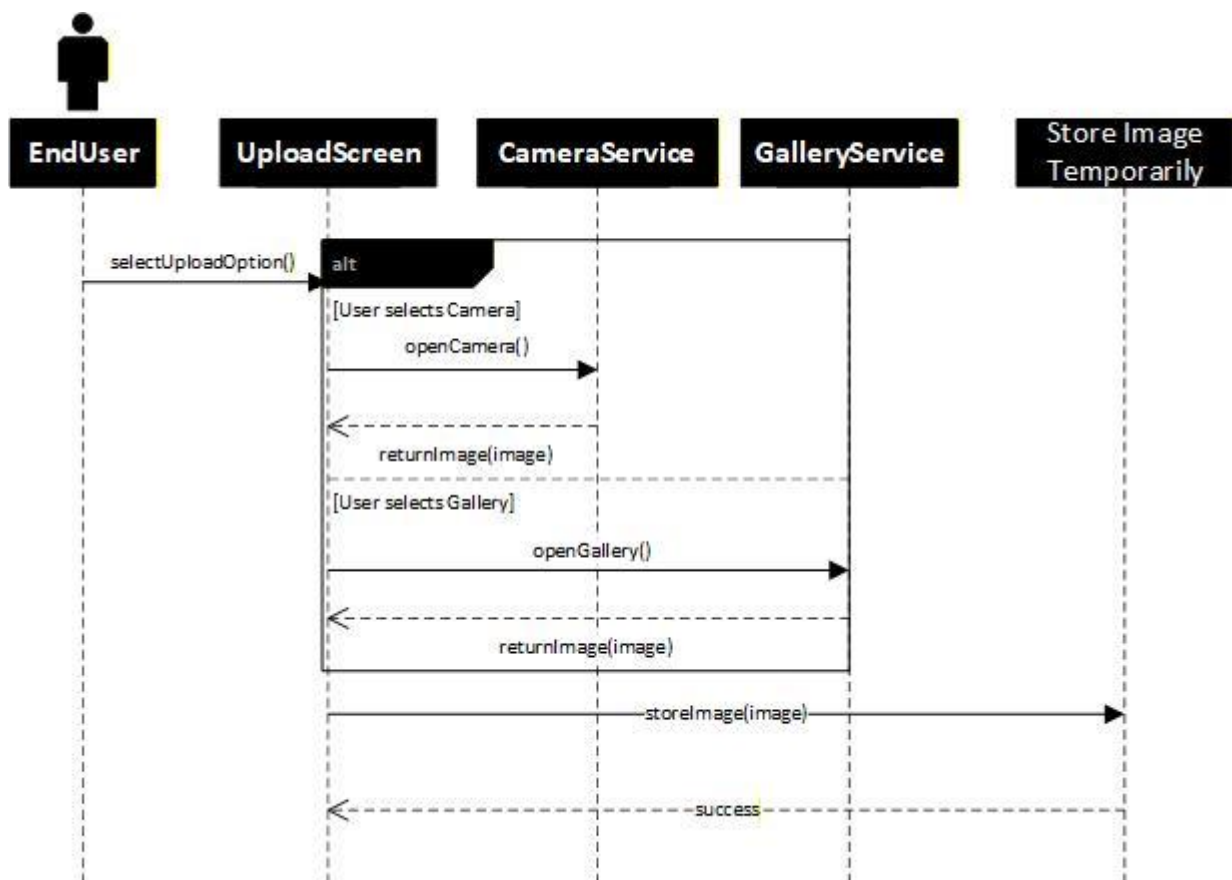


Figure 4 Sequence Diagram for Upload Image (UC2)

This sequence diagram illustrates the process of generating personalized dress recommendations. When the user selects the recommendation option, the system retrieves the latest body measurements from the MeasurementService. The RecommendationEngine processes these measurements and matches them with available dress sizes stored in the catalog database. The recommended dresses are then displayed to the user. Alternative flows such as no matching results or database failures are also handled.

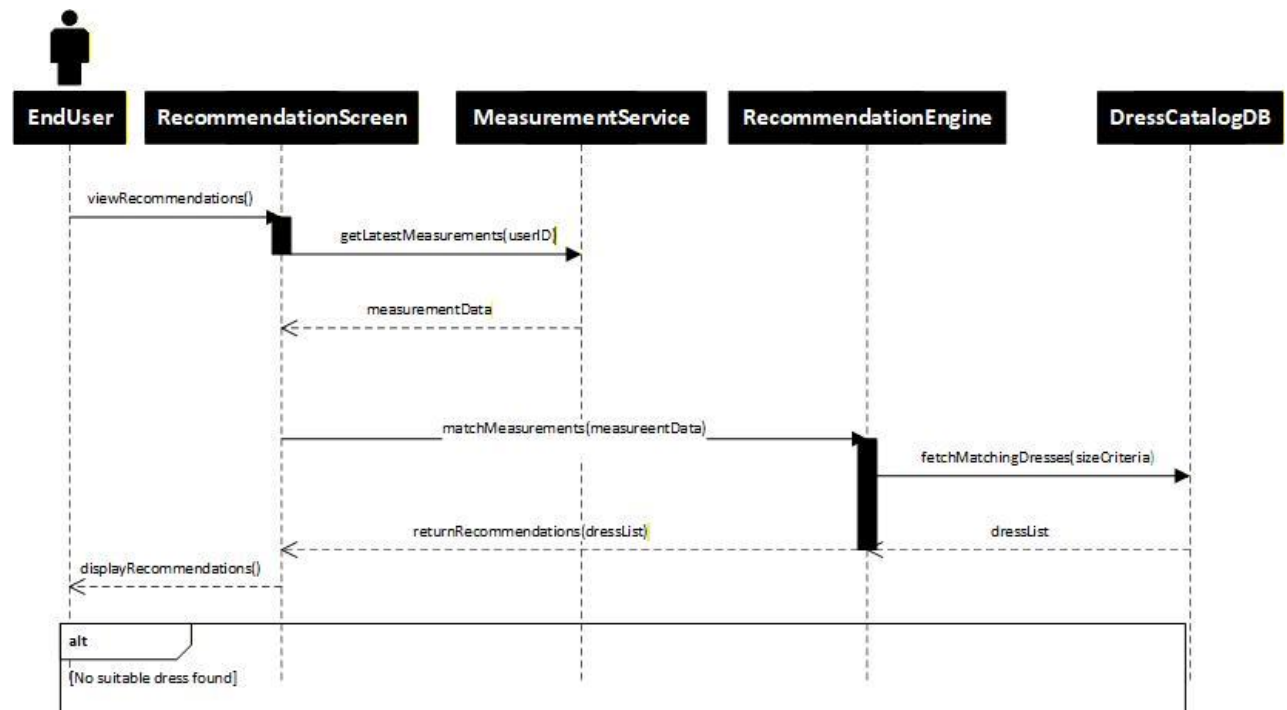


Figure 5 Sequence Diagram for View Dress Recommendations (UC3)

This sequence diagram represents the process of saving body measurements for future reference. After measurements are generated, the user selects the save option. The system validates the measurement data and stores it securely in the Firestore database. Upon successful storage, a confirmation message is displayed. The diagram also illustrates alternative scenarios such as user cancellation or database connectivity failure.

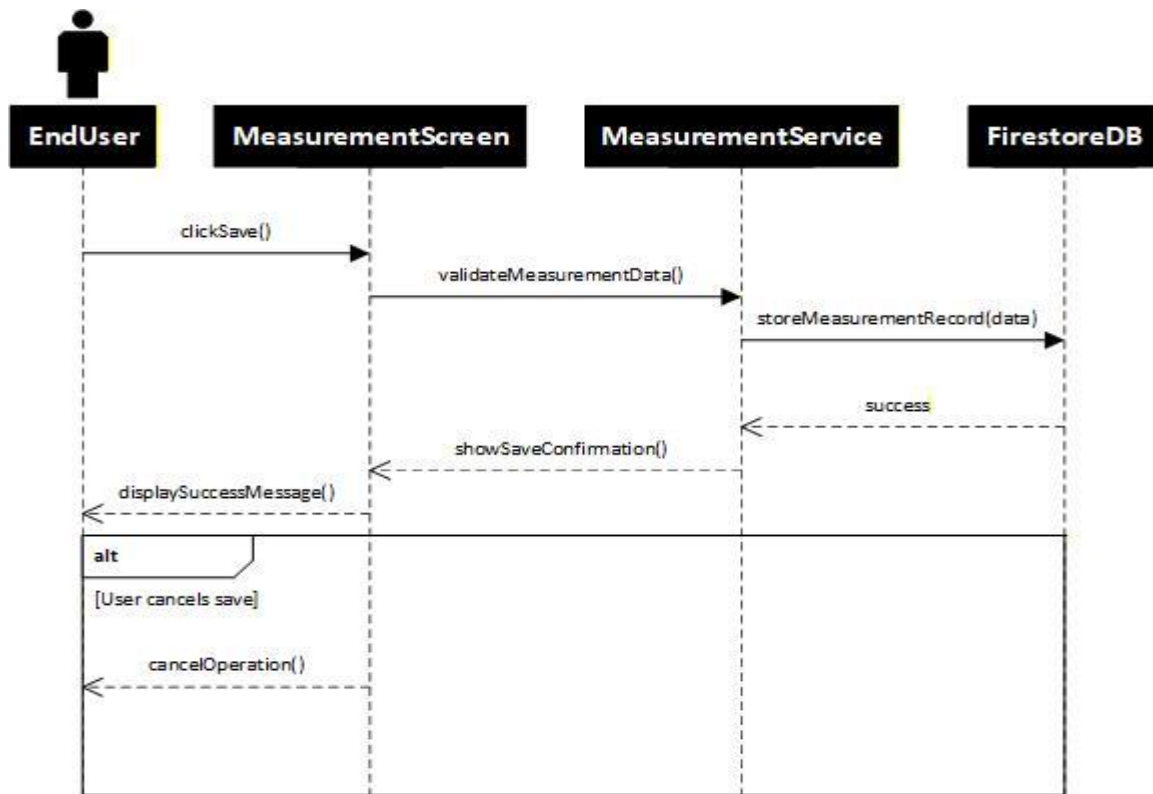


Figure 6 Sequence Diagram for Save Measurement History (UC4)

This sequence diagram illustrates the process of viewing previously saved body measurements. When the user selects the view history option, the system retrieves stored measurement records from the Firestore database through the MeasurementService. The retrieved records are then displayed to the user. The diagram also handles alternate scenarios such as no available history or database access errors.

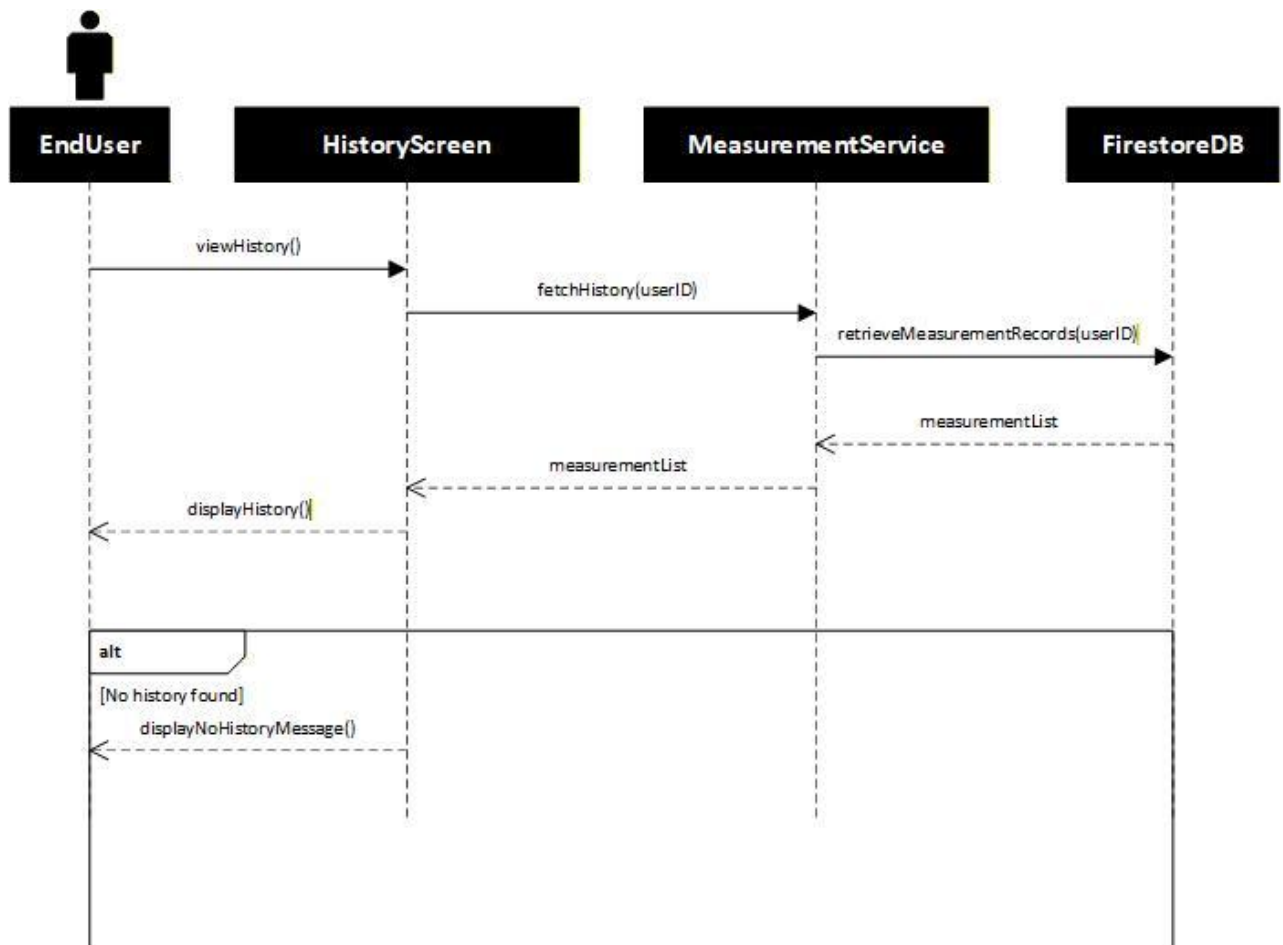


Figure 7 Sequence Diagram for View Measurement History (UC5)

This sequence diagram illustrates how the system handles notifications and alerts. When a system event occurs, the SystemEventManager triggers the NotificationService to generate a notification. The notification is then delivered to the user's device and displayed as an alert. The diagram also represents alternative scenarios such as disabled notifications or delivery failure.

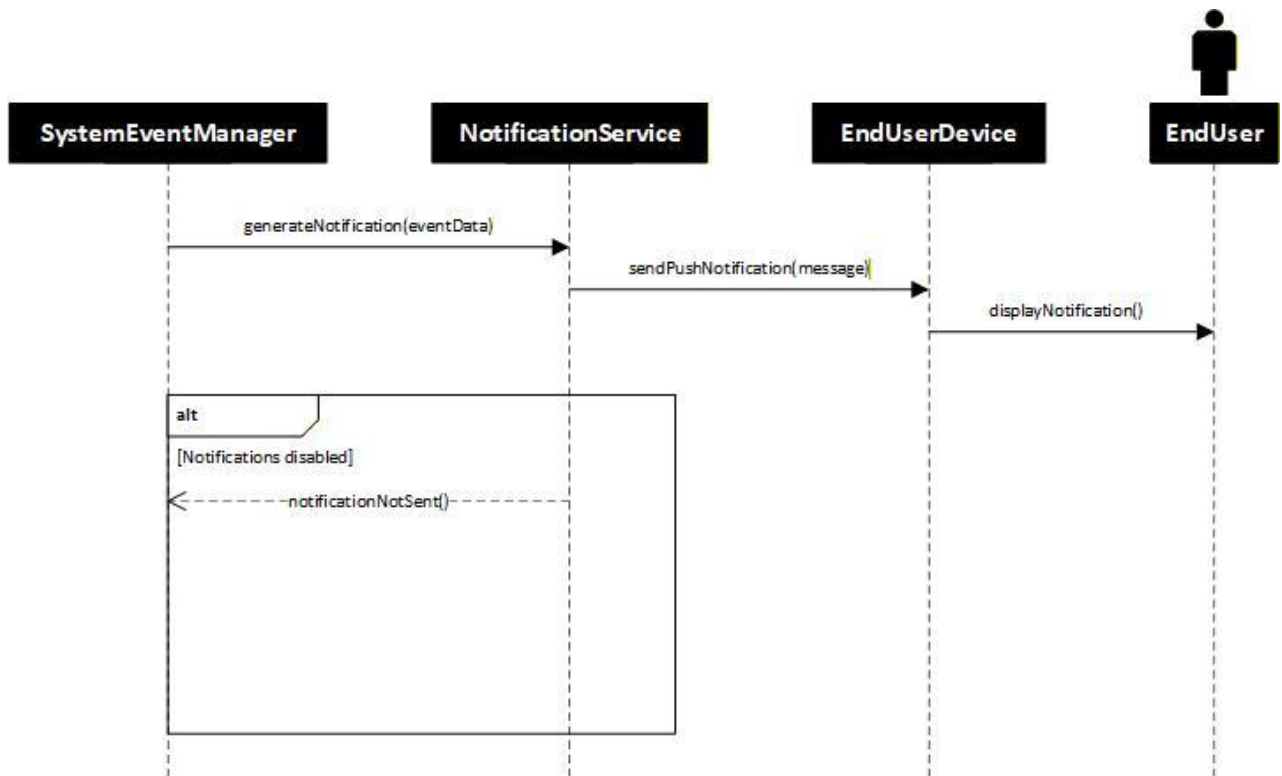


Figure 8 Sequence Diagram for Notifications and Alerts (UC7)

3. State Transition Diagram

The State Transition Diagram describes the logical states of the AI-Based Body Measurement App and the transitions between these states during execution. It presents how the system changes state in response to user actions and internal events.

Typical states include Idle, Image Capture, Pose Detection, Measurement Calculation, Result Display, and Data Storage. Transitions occur when events such as image capture completion, successful pose detection, or measurement calculation are triggered. Additional states include Save Measurement and Process Completed, representing persistent storage and completion of the measurement workflow.

This diagram provides a design-level behavioral view of the system and helps developers understand valid system states and transitions, reducing logical errors during implementation.

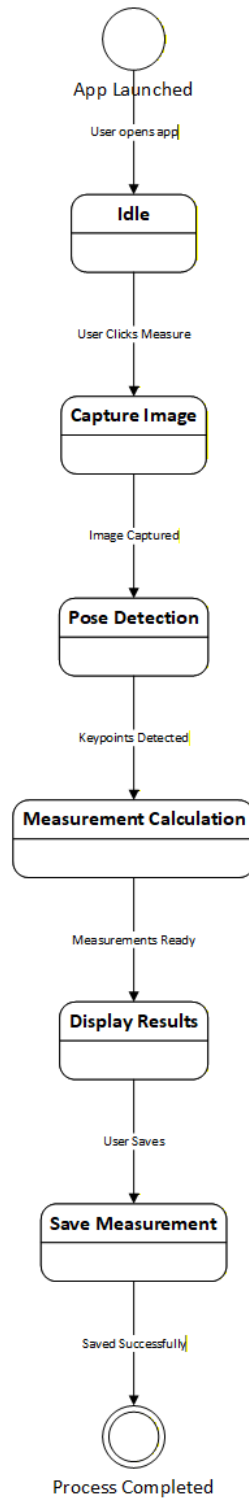


Figure 9 State Transition Diagram

4.5 Architectural Design

The AI-Based Body Measurement App follows a Layered Client–Server Architecture, ensuring clear separation of concerns, scalability, and ease of maintenance.

- The Presentation Layer manages user interactions such as login, image capture, and result display using a mobile UI framework.
- The Application Logic Layer controls workflow execution, coordinating between UI, AI processing, and data storage.
- The AI Processing Layer performs on-device pose detection using MoveNet / ML Kit and calculates body measurements.
- The Data Layer manages authentication and persistent storage using Firebase Authentication and Firebase Firestore / Realtime Database.

This architecture ensures on-device processing for privacy, efficient communication between components, and flexibility for future enhancements.

1. UML Component diagram

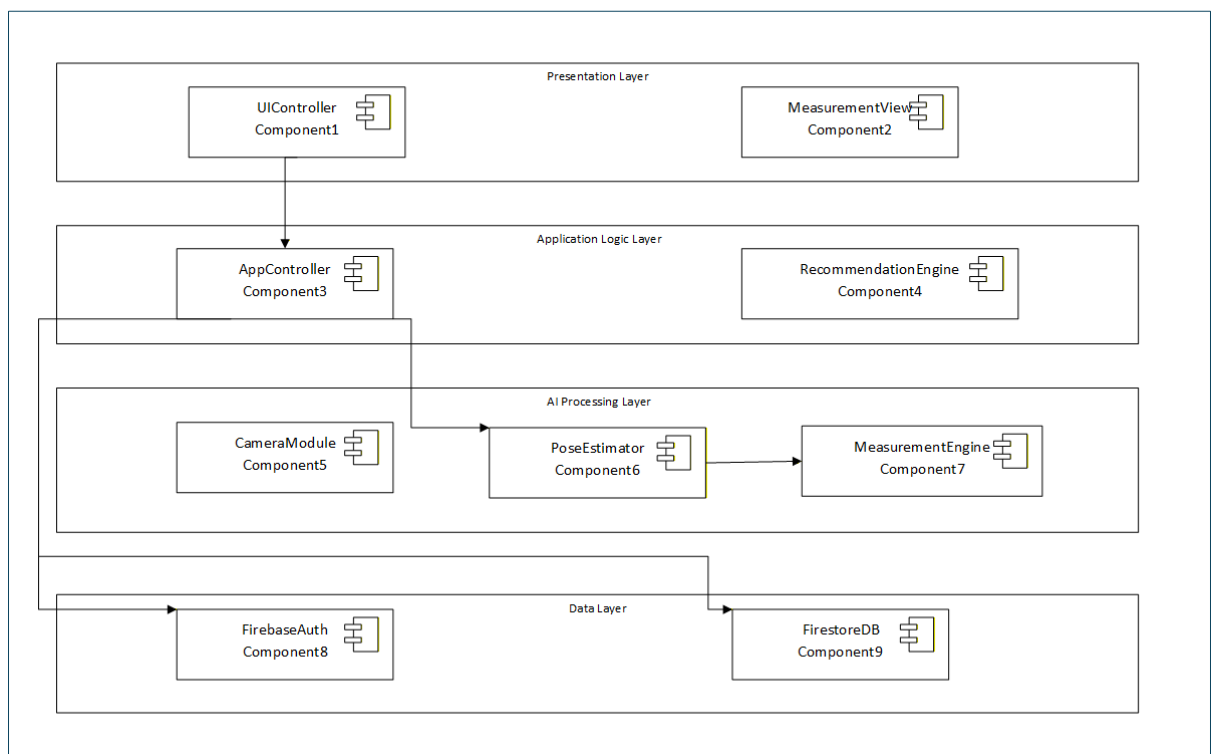


Figure 10 Layered and Client–Server Architectural Design.

4.6 Data Design

The Data Design defines how data is structured, stored, and managed within the AI-Based Body Measurement App. The primary entities include User, MeasurementRecord, and UserAuth.

User profiles and measurement data are stored in Firebase Firestore / Realtime Database using a NoSQL structure for flexibility and scalability. Each measurement record includes a unique ID, timestamp, and calculated body dimensions. Relationships between users and measurements are maintained using user identifiers.

This design ensures efficient data retrieval, secure storage, and support for historical measurement tracking.

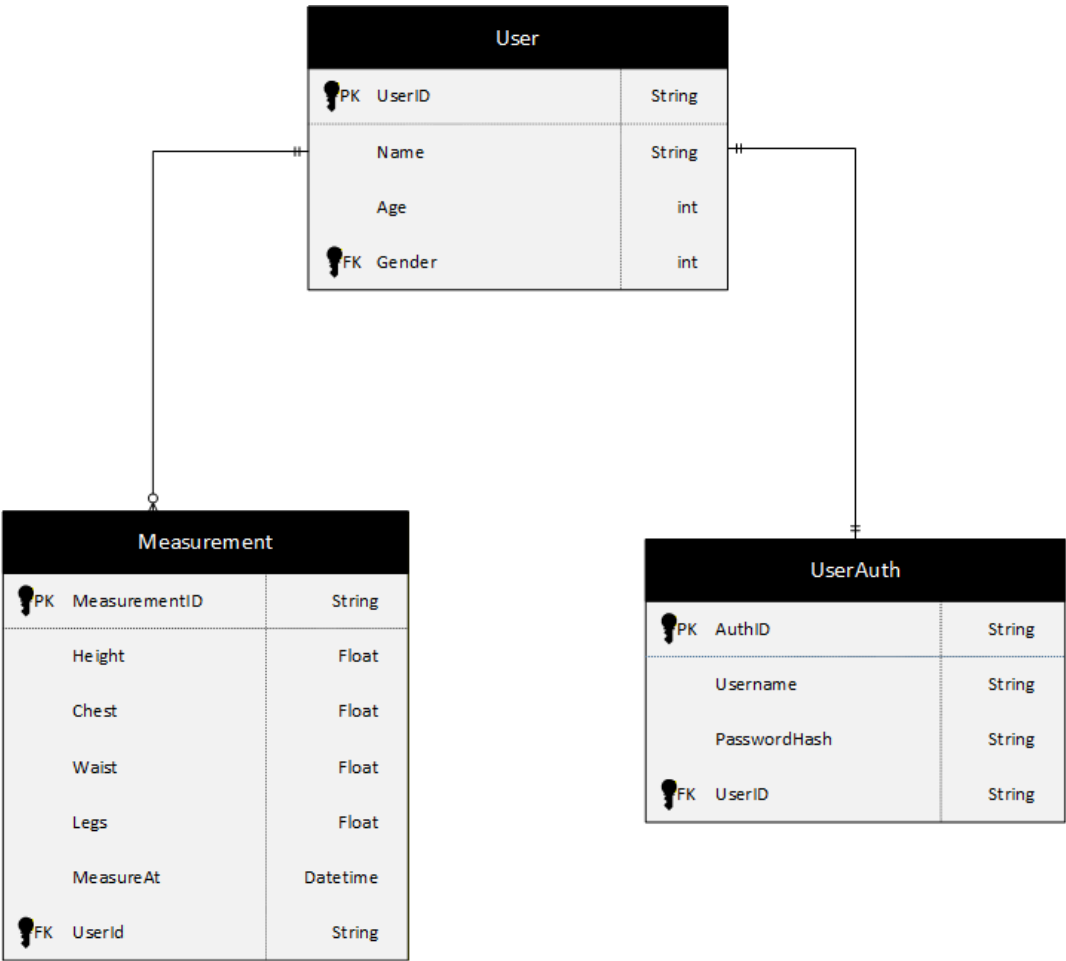


Figure 11 Data Design (ERD) Diagram

1. Data Dictionary

Table 21 Alphabetical List of Sistem Entities Data Dictionary

Terminology	Description
Age	Integer. Represents the age of the user, stored as part of the user profile.
Chest	Float. Stores the chest measurement calculated by the AI measurement engine.
Confidence	Float. Indicates the confidence level of each detected pose landmark generated by MoveNet.
Coordinates	String / Array. Stores the X and Y coordinates of body landmarks detected during pose estimation.
Email	String. User's email address used for login and identification purposes.
Gender	String. Represents the gender of the user for profile completeness.
Height	Float. Stores the calculated height measurement of the user.
LandmarkID	String. Unique identifier for each pose landmark.
MeasurementID	String. Unique identifier assigned to each measurement record.
MeasurementRecord	Object. Stores calculated body measurements such as height, chest, waist, and hip.
Name	String. Stores the full name of the user.
Password	String. Encrypted password used for user authentication.
PoseLandmark	Object. Represents detected body joints used for measurement calculations.
SessionID	String. Unique identifier representing a measurement session or history entry.
Timestamp	DateTime. Records the date and time when a measurement is taken.
User	Object. Represents a registered user of the application.
UserID	String. Unique identifier assigned to each user.
Waist	Float. Stores the waist measurement calculated by the AI module.

• Structured Approach: Functions and Function Parameters:

Table 22 Structured Approach Functions and Function Parameters

Function Name	Description	Parameter Name	Data Type	Parameter Role
captureImage()	Captures an image or video frame from the device	cameraFrame	ImageFrame	Stores the captured image from the device camera.

	camera for pose detection.			
detectPose()	Detects body pose landmarks using the MoveNet model.	imageFrame	ImageFrame	Provides the input image for pose detection.
calculateMeasurements()	Calculates body measurements such as height, chest, waist, and hip using ML Kit.	poseLandmarks	List<PoseLandmark>	Contains detected body landmarks used for measurement calculation.
saveMeasurement()	Saves the calculated body measurements to the Firebase database.	measurementRecord	MeasurementRecord	Holds measurement values and related metadata for storage.
getMeasurementHistory()	Retrieves previous measurement sessions for a specific user.	userID	String	Identifies the user whose measurement history is requested.
authenticateUser()	Authenticates the user during login.	email	String	Used to identify the user account.
authenticateUser()	Authenticates the user during login.	password	String	Used to verify user credentials securely.
createUserProfile()	Creates a new user profile in the system.	userData	User	Contains user information such as name, email, age, and gender.

- **Object-Oriented (OO) Approach: Objects, Attributes, Methods, & Method Parameters:**

- User Object:

Element	Details
Attributes	UserID (String), Name (String), Email (String), Age (Integer), Gender (String), Password (String)
Description	Represents a registered user of the application and stores profile and authentication information.
Methods	registerUser(), loginUser()
Method Parameters	registerUser(name:String, email:String, password:String, age:Integer, gender:String)loginUser(email:String, password:String)

- MeasurementRecord Object

Element	Details
Attributes	MeasurementID (String), Height (Float), Chest (Float), Waist (Float), Hip (Float), Timestamp (DateTime)
Description	Stores body measurement values calculated by the AI module.
Methods	calculateMeasurements(), saveMeasurement()
Method Parameters	calculateMeasurements(poseLandmarks:List<PoseLandmark>)saveMeasurement()

- PoseLandmark Object

Element	Details
Attributes	LandmarkID (String), Coordinates (String/Array), Confidence (Float)
Description	Represents detected body joints generated by the MoveNet pose detection model.
Methods	detectLandmark()
Method Parameters	detectLandmark(imageFrame:ImageFrame)

- CameraHandler Object

Element	Details
Attributes	Resolution (String), CameraStatus (Boolean)
Description	Handles camera operations for capturing images used in pose detection.
Methods	captureImage()
Method Parameters	captureImage(): ImageFrame

4.7 User Interface Design

The AI-Based Body Measurement App allows users to measure their body dimensions easily using a mobile phone camera. The system is designed to be simple so that any user can use it without technical knowledge.

When the user opens the app, they first register or log in using their email and password. After logging in, the user sees the main screen, where they can start a new measurement or view their previous measurement records.

To take a measurement, the user opens the camera and follows simple on-screen instructions to stand in the correct position. The system captures the image and uses AI pose detection (MoveNet) to detect body points. These points are then used by the measurement module (ML Kit) to calculate body measurements such as height, chest, waist, and hip.

After processing, the calculated measurements are shown clearly on the screen. The user also receives feedback messages, such as confirmation of successful measurement or guidance to adjust posture if needed. All measurement results are automatically saved so the user can access them later.

The user can view all past measurements in the history section, where results are displayed with date and time. This helps the user track body changes over time.

Overall, the system provides an easy, accurate, and user-friendly way to measure the body, view results instantly, and maintain a history of measurements.

1. Screen Images

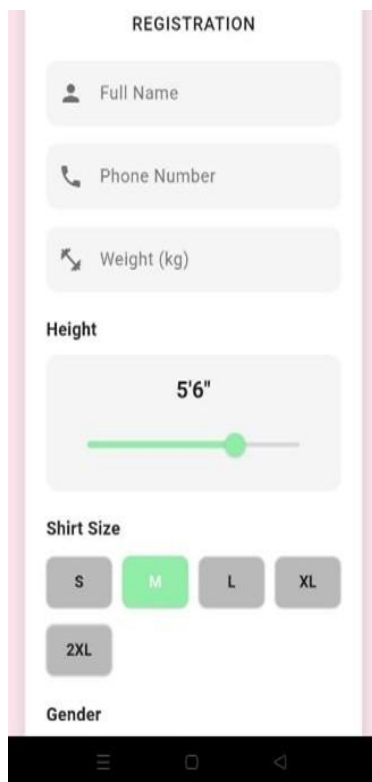


Figure 12 User Registration

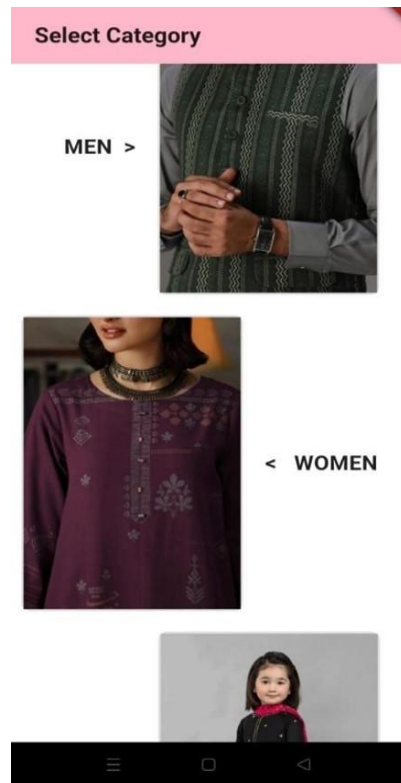


Figure 13 User select Category

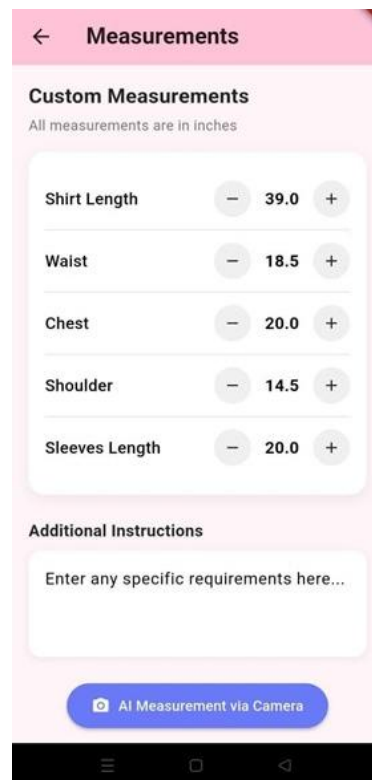


Figure 14 Manual Measurements



Figure 15 User clicking Image

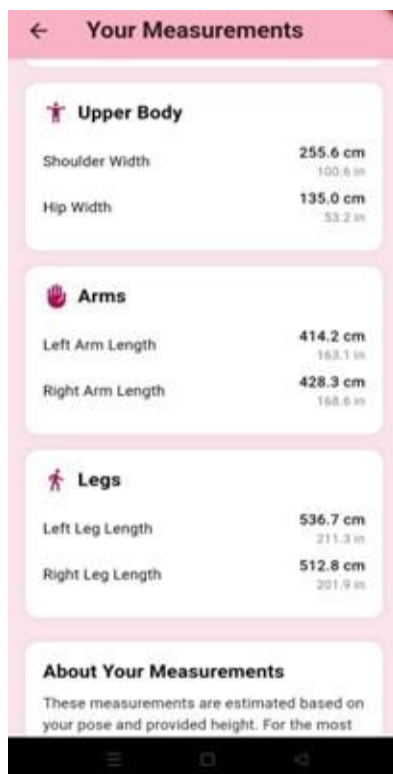


Figure 16 Displaying Result



Figure 17 Recommended Dresses

2. Screen Objects and Actions

Screen Object	Description	Action Performed
Capture Button	Button used to initiate image capture	Activates the device camera to capture body image
Retake Button	Allows user to discard current image	Clears captured image and reopens camera
Confirm Button	Confirms captured image	Sends image to pose detection and measurement module
Back Button	Navigation control	Returns user to home screen

Screen Object	Description	Action Performed
Measurement Result Panel	Displays calculated body measurements	Shows height, width, and other computed values
Recommendation Button	Triggers recommendation process	Generates clothing size and design suggestions
Save Button	Stores current results	Saves measurements and recommendations in database

4.8 Design Decisions

During the design and development of the AI-Based Body Measurement App, several important architectural and technical decisions were made. These decisions were not random; rather, they were carefully evaluated based on system requirements such as accuracy, scalability, performance, security, and maintainability.

This section highlights the major design choices, the alternative approaches that could have been used, and the justification for selecting the final approach.

Design Choice	Approach Used	Alternative Considered	Reason for Selection	Benefit
Software Design Paradigm	Object-Oriented Design (OOD)	Procedural Programming	Better modeling of real-world entities such as User, PoseLandmark, and MeasurementRecord	Improved modularity, maintainability, and reusability
System Architecture	Layered Architecture + Client-Server Model	Monolithic Architecture	Clear separation between UI, AI processing, business logic, and database layers	Better scalability, debugging, and security
AI Model Selection	MoveNet (TensorFlow Lite)	MediaPipe Pose	Lightweight, optimized for mobile devices, real-time inference	High speed and on-device accuracy
Vision Processing Support	ML Kit	Custom image processing algorithms	Ready-made mobile vision utilities with optimized performance	Faster development and reliable processing
Database Design	NoSQL (Firebase Firestore)	Relational SQL Database	Flexible schema for dynamic measurement records	Real-time sync, scalability, offline support
Measurement Algorithm	Geometric distance-based computation using body keypoints	Manual input or hardware sensors	Fully automated measurement from detected pose landmarks	Reduced human error and consistent results
UI Framework	Flutter	Native Android	Single codebase, easy Firebase and AI integration	Faster development and cross-platform capability

4.9 Summary

In this chapter, the key design ideas of the AI-Based Body Measurement App were discussed and refined to ensure the system meets its objectives efficiently and accurately. The object-oriented design was chosen to model real-world entities, while the layered client-server architecture ensures clear separation of concerns among the user interface, application logic, AI processing, and data storage.

Critical design decisions, including the use of MoveNet for pose detection, ML Kit for measurement calculation, and Firebase for flexible database management, were made to maximize accuracy, speed, and user privacy. The chapter also highlighted algorithmic approaches for body measurement calculation, UI design choices using Flutter, and component interactions to ensure smooth workflow and maintainability.

Overall, the design decisions and refinements presented in this chapter provide a solid foundation for implementing the system, ensuring that all project objectives—accurate body measurement, user-friendly interface, data management, and maintainability—are fully addressed.

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